ChatGPT

Designing and Prioritizing Memory-Dependent / State-Consistent Calibration for Dynamic-Memory Industrial Anomaly Detection

Executive summary

The strongest research opportunity now is **not another memory-admission mechanism**. It is to make the operating threshold consistent with a detector whose scoring function changes whenever its Greedy normal memory changes.

Your recent experiments make that problem concrete. The P0/P1 protocol used the same frozen four-shot source bank, the same v22 normal-patch generator and Greedy update operator, 128 causal queries yielding 85 verified normals and 43 queried defects, and an additional 64 normal images reserved for calibration. It explicitly kept Probe B untouched and prohibited Probe A from influencing non-oracle threshold or admission decisions. The existing repository also already contains the extreme-order experiments—Normal-first, Defect-first, native/random-like streams—and R6 shows that temporal ordering can drive the same online system to very different FPR/FNR operating regimes. For example, `switch`, support seed 0, had FNR 0.722 under Normal-first but 0.090 under Defect-first, while FPR moved in the opposite direction.

The key mathematical issue is simple. With frozen DINOv2 features but changing normal memory,

$$s_t(x) = S\left(f(x); M_{\text{fix}}, M_t^{\text{dyn}}\right)$$

is a **time-varying score function even though f is frozen**. A score cached when an anchor was labelled at t_i,

$$s_{t_i}(x_i),$$

is generally not the score that the current detector assigns to that same anchor,

$$s_t(x_i).$$

Using both interchangeably in one threshold controller mixes measurements made under different detectors.

This observation is well grounded in external literature—but that is also a warning about novelty. Gibbs and Candès explicitly formulate online conformal inference with a time-varying score function S_t and corresponding time-varying quantile function $\widehat{Q}_t$, noting that both may need to be re-estimated as the environment changes. Their ACI update then adjusts the calibration level according to realized errors. [1] Sun, Sankararaman and Narayanaswamy go even closer to our problem: their ICML 2024 paper studies

online anomaly threshold selection under distribution shifts using confidence sequences, including offline warm starts and FPR/FNR control in their model. [2] Therefore, "recompute a quantile after the model changes" alone is **not enough novelty**.

The potentially publishable SCDAC contribution is the combination of four things that those methods do not directly solve in your setting:

> **versioned current-memory rescoring + leave-source-out anti-self-matching + explicit two-sided evidence states + selective-feedback-aware threshold control.**

That combination directly addresses your particular endogenous drift: the *detector itself* moves because verified normal examples modify its retrieval memory, while labels arrive sparsely and non-uniformly.

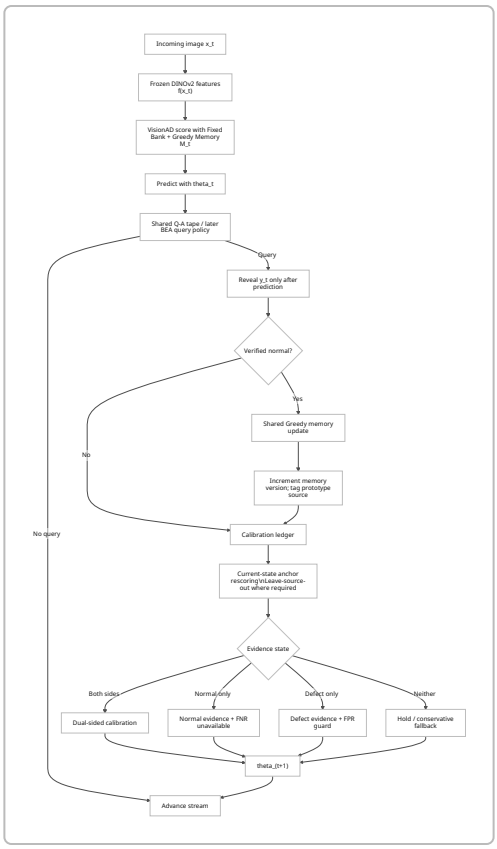
My algorithm priority is therefore:

Priority	Candidate	Research role	Recommendation
A	**State-Consistent Quantile, SCQ**	strongest simple falsification baseline	implement first
B	**Guarded SCDAC, SCDAC-G**	best near-term paper-centerpiece candidate	implement immediately after SCQ
C	**State-Consistent Dual-ACI / OCP**	strongest literature-derived adaptive competitor	implement in the same first comparison
D	**Confidence-Sequence SCDAC**	theory-oriented extension	only after B survives
E	**Propensity-Aware SCDAC**	handles BEA/selective-label bias	second-stage extension
F	**Covariate-Weighted SCDAC**	distinguishes environmental shift from memory-induced score drift	only if temporal drift remains the dominant failure

For the fastest credible paper path, I would **not implement D–F initially**. First build a common calibration ledger and evaluate **SCQ, SCDAC-G, ACI, SPOT, original Sidecar and frozen threshold** under an identical Greedy memory path and identical Q-A tape. If SCDAC-G cannot beat SCQ and adapted ACI across classes/seeds/orders, it should not become the paper centerpiece.

The proposed controller should also refuse to confuse *absence of evidence* with *zero risk*. If no useful defect evidence exists during a long Normal-first interval, the correct state is `FNR_UNAVAILABLE`, not $\widehat{\mathrm{FNR}} = 0$. Conversely, if normal and defect evidence imply mutually incompatible thresholds, the correct state is `RISK_CONSTRAINT_CONFLICT`, not an apparently precise but unjustified threshold.

The recommended architecture is:



The central recommendation is therefore:

Treat SCQ as the falsifier, SCDAC-G as the principal candidate, and adapted ACI plus Sun/SPOT as the external challenge. Do not add propensity, confidence-sequence or covariate-weighting machinery until the simple state-consistent mechanism has shown a reproducible advantage.

Research problem and constraints

Formal problem

At time t, let

$$z_t = f(x_t)$$

be the frozen DINOv2 representation. The deployed anomaly scorer is

$$s_t(x) = S(f(x); M_{\text{fix}}, M_t)\,,$$

where M_{fix} is the immutable support-derived bank and M_t is the dynamic Greedy normal coreset.

The binary prediction is

$$\hat{y}_t = \mathbf{1}[s_t(x_t) \geq \theta_t].$$

The query decision must be made from information available before seeing y_t:

$$q_t = \pi_q(H_{t^-}, s_t, z_t, \theta_t),$$

and the label is revealed only if $q_t = 1$. Only after the current prediction and query decision may verified feedback change either memory or the threshold.

A strict event sequence should therefore be

$$\boxed{\text{score} \to \text{predict} \to \text{query} \to \text{reveal label} \to \text{update memory} \to \text{rescore calibration state} \to \text{update } \theta_{t+1}}$$

and **never**

$$\text{reveal current label} \to \text{change } \theta_t \to \text{retroactively classify current sample}.$$

The current repository already contains the relevant frozen-v22 architecture, AIF-inspired query choices and guarded Sidecar/rescue machinery; the v22 source explicitly records the addition of guarded dual-anchor thresholding, warm-up/audit-query behavior and balanced rescue.

The two different kinds of drift

SCDAC should separate two phenomena that are easy to conflate.

Endogenous score drift occurs because the detector changes:

$$M_t \to M_{t+1} \quad \Rightarrow \quad S(\cdot; M_t) \;/\; S(\cdot; M_{t+1}).$$

Environmental distribution drift occurs because

$$P_t(X, Y) \;/\; P_{t+1}(X, Y).$$

Current-state rescoring directly fixes the first problem. It does **not** automatically solve the second. This distinction is crucial because weighted conformal work under covariate shift requires conditions such as stability of the conditional response mechanism and knowledge or accurate estimation of a covariate likelihood ratio; those assumptions cannot simply be asserted for an industrial stream. [3]

A third problem: self-calibration through memory

There is an additional issue specific to memory methods.

Suppose verified normal image x_i contributes prototypes $P_i \subset M_t$. If we later use x_i as a calibration anchor and score it against a memory containing its own prototypes, then

$$S(x_i; M_t)$$

may be artificially small.

Therefore, for any normal anchor that has contributed to dynamic memory, I recommend defining a source-aware score

$$\boxed{\tilde{s}_{i,t} = S(x_i; M_{\text{fix}}, M_t \setminus P_i)}$$

whereas a protected calibration-only normal that was never written to memory can simply use

$$\tilde{s}_{i,t} = S(x_i; M_{\text{fix}}, M_t).$$

This **leave-source-out current-state rescoring** is, in my view, one of the most valuable additions to SCDAC. It prevents state consistency from reintroducing the exact self-matching issue that affected earlier threshold initialization.

What the recent repo evidence implies

P0/P1 already establishes the information-access discipline we should preserve: four-shot fixed source support; 64 extra normal calibration images explicitly disclosed; no Probe B access; no Probe A access for non-oracle threshold or admission selection; and a frozen query tape whose selection does not read ground truth.

It also records 128 queries with 85 verified normals and 43 defects and uses the same native v22 normal-patch and Greedy operator across compared memory controls. The output archive includes calibration scores, query tape, admission decisions, Probe-A scores and provenance/checksum files, giving us most of the infrastructure required for reproducible calibration experiments.

The extreme-order R6 audit makes temporal imbalance impossible to dismiss as a corner case. In `switch` seed 0, the existing balanced controller reached BA 0.638 under Normal-first versus 0.897 under Defect-first, with the former showing FNR 0.722 and the latter FNR 0.090. The same artifact explicitly records windows with no normal or no defect examples.

That leads to six hard constraints for the new method:

Constraint	Consequence for SCDAC
Frozen DINOv2 / VisionAD representation	adaptation must occur outside backbone gradients
Greedy dynamic normal memory	calibration scorer itself changes after verified-normal writes
$B \in \{16, 32, 64, 79, 128\}$	defect evidence can be extremely sparse at low B
Strict causal query protocol	no current/future label may alter current prediction
Extreme Normal-first / Defect-first / Random / Bursty order	one side of class-conditional risk may disappear for long periods
Selective feedback	queried error rates need not equal population error rates

The main method should therefore be designed not merely to “adapt a threshold,” but to answer:

> **What threshold is defensible given the current memory state, the evidence that is actually available, and the sampling mechanism that produced that evidence?**

Literature and baseline hierarchy

The external literature is rich enough that lack of baselines should not be a problem this time. The more important challenge is choosing baselines that test distinct failure hypotheses instead of implementing every online-calibration paper.

Highest-priority calibration references

Priority	Method / source	One-line contribution	Why it matters here
Must	**Sun et al., ICML 2024 — Online Adaptive Anomaly Thresholding with Confidence Sequences**	Directly studies online anomaly thresholds under distribution shifts, confidence sequences, FPR/FNR control, and warm starts. [2]	Closest external competitor to the paper-level problem statement.
Must	**Gibbs & Candès, NeurIPS 2021 — ACI**	Uses an online update $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \mathrm{err}_t)$ to adapt calibration under arbitrary distribution sequences in its prediction-set setting. [4]	Strong, simple adaptive baseline; also shows that “time-varying score needs time-varying calibration” is not by itself novel.
Must	**Gibbs & Candès, JMLR 2024**	Extends ACI by adapting its step size and obtains small regret on local time intervals. [5]	Particularly relevant to Bursty and changing-rate shifts.
Must	**Siffer et al., KDD 2017 — SPOT**	Uses extreme-value theory to set streaming anomaly thresholds from score tails with a risk parameter rather than a manually chosen cutoff. [6]	Label-free streaming threshold comparator; tests whether SCDAC’s labels/state machinery is actually necessary.
Must	**Rolling / empirical quantile**	Maintain a recent score quantile as the threshold.	Simplest credible adaptive threshold and an essential falsifier.
Strong	**Angelopoulos, Barber & Bates, ICML 2024**	Online conformal calibration with decaying step sizes; maintains arbitrary-sequence retrospective coverage while improving stable-distribution pointwise behavior. [7]	Modern alternative to fixed-step ACI.

Priority	Method / source	One-line contribution	Why it matters here
Strong	**Areces et al., ICML 2025 — OCP via online optimization**	Gives online conformal algorithms for adversarial and stochastic data and reports stronger gains as temporal dependence increases. [8]	Especially relevant if ACI is competitive and temporal dependence is important.

ACI deserves special attention. Its paper already allows a fitted score function S_t and corresponding quantile function $\widehat{Q}_t$ to change over time and explicitly says these should be re-estimated to align with recent observations. [1] Thus, a new SCDAC paper should **not** claim that the abstract idea of state-dependent calibration is unprecedented. The novelty must lie in the particular challenges of memory-governed scoring, source leakage, sparse class-conditional evidence and selective feedback.

Covariate shift and selective-label references

Weighted conformal prediction under covariate shift shows that calibration samples can be reweighted when train and test covariate distributions differ, provided an appropriate likelihood ratio is known or accurately estimated and the relevant conditional mechanism remains stable. [3] This makes it a natural conceptual foundation for a later covariate-weighted SCDAC, but not the first algorithm I would implement.

Selective labels are a genuine statistical concern: Lakkaraju et al. show that evaluating predictors on selectively observed outcomes can yield erroneous conclusions because the observed labels need not represent the population. [9] Importance-Weighted Active Learning explicitly uses importance weights to correct active-sampling bias while controlling estimator variance. [10]

Two newer references are even more directly relevant to the future BEA integration. Xu, Karampatziakis and Mineiro develop active, anytime-valid risk-controlling prediction sets under adaptively collected labels and explicit label budgets. [11] CAP studies online conformal inference after adaptive selection and builds selection-aware calibration sets with finite-sample guarantees in its setting. [12] Neither should be treated as a drop-in proof for SCDAC, but both establish that **adaptive selection must be part of the statistical model**, not ignored.

Query-policy references

Calibration should initially be tested under the frozen Q-A tape so query behavior cannot confound threshold behavior. Only after a calibrator is frozen should it be combined with BEA.

For that second stage, there are credible external query-policy references. Confidence-Budget Matching queries feedback when uncertainty is large relative to remaining budget and is explicitly designed for hard feedback budgets. [13] Cai and Kandasamy study cost-aware active labeling of streaming data with uncertainty- and budget-dependent querying and analyze adaptation to different arrival patterns. [14] Agarwal's selective-sampling work provides a streaming, cost-sensitive active-learning comparator with regret and label-complexity analysis. [15] Importance-weighted active learning is the natural statistical reference if we deliberately randomize query decisions and record positive propensities. [10]

The repository's own v22 code already exposes AIF-inspired and simpler query modes, so Random, Periodic, Margin and Novelty remain necessary internal controls even if BEA is eventually restored.

Baseline taxonomy

The important distinction is that some requested comparators alter **calibration**, while others alter **memory**.

Baseline	Changes memory?	Changes threshold?	Uses labels?	Primary purpose
Frozen θ_0	no	no	initialization only	no-adaptation control
Rolling Quantile	no	yes	optional	simplest adaptive calibration
SPOT	no	yes	no	streaming unlabeled threshold
Original Sidecar	no	yes	yes	internal historical controller
ACI adaptation	no	yes	queried labels	external adaptive calibration
Sun-CS adaptation	no	yes / abstain	method-dependent	strong external anomaly-threshold comparator
Greedy + state-consistent q90	Greedy only	yes	64 calibration normals	strongest simple memory+calibration baseline
Causal Novelty	yes	optionally same calibrator	queried normal labels	deployable memory-policy comparator
Novelty Top-K	yes	optionally same calibrator	hindsight admission ranking	**retrospective diagnostic**, not deployable
R1 Beam Oracle replay	yes	optionally same calibrator	Probe-A hindsight	**oracle diagnostic**, not deployable

The P0/P1 artifact itself explicitly marks its exact Top-K controls as retrospective and the R1 oracle as Probe-A-label selected. They therefore should never appear in the same "online baseline" column as ACI or SPOT without a clear oracle/retrospective label.

Algorithmic design requirements

A good SCDAC method needs more than a smarter objective function.

State consistency and source independence

Every calibration record should be a **versioned anchor**, not merely a scalar score:

$$a_i = (\mathrm{id}_i, y_i, z_i, t_i, v_i, p_i, \mathcal{P}_i),$$

where v_i is the memory version at acquisition, p_i is query propensity if known, and $\mathcal{P}_i$ identifies prototypes contributed by that sample.

After memory version v_t changes, any calibration quantity used for θ_{t+1} should be obtained from the current scorer:

$$r_{i,t} = \begin{cases} S(z_i; M_t \setminus \mathcal{P}_i), & y_i = 0,\ \mathcal{P}_i\ / \\ S(z_i; M_t), & \text{otherwise.} \end{cases} = \varnothing,$$

The old value $s_{t_i}(x_i)$ remains useful only as a **drift diagnostic**.

Evidence must have explicit states

The controller should expose:

$$E_t \in \{\texttt{BOTH}, \texttt{NORMAL_ONLY}, \texttt{DEFECT_ONLY}, \texttt{NONE}, \texttt{CONFLICT}\}.$$

`NORMAL_ONLY` must not imply FNR $= 0$; `DEFECT_ONLY` must not imply FPR $= 0$.

That semantic distinction matters directly in extreme streams. The R6 artifact contains numerous windows without one class and shows strong order-dependent FNR/FPR shifts.

Two-sided threshold geometry

For anomaly scores where larger means more anomalous, define class-conditional risks

$$R_{\mathrm{FP},t}(\theta) = P_t(s_t(X) \geq \theta \mid Y = 0),$$
$$R_{\mathrm{FN},t}(\theta) = P_t(s_t(X) < \theta \mid Y = 1).$$

Increasing θ tends to reduce FPR and increase FNR.

If one chooses explicit risk targets α and β, normal evidence induces approximately

$$\theta \geq q_t^N(1 - \alpha),$$

while defect evidence induces

$$\theta \leq q_t^D(\beta).$$

Therefore a two-sided feasible set exists only when

$$q_t^N(1-\alpha) \le q_t^D(\beta).$$

With uncertainty, a conservative analogue is

$$U[q_t^N(1-\alpha)] \le \theta \le L[q_t^D(\beta)].$$

If the interval is empty, the correct result is not “optimize harder.” It is evidence that the requested operating point is incompatible with the currently observed score separation.

For the paper, however, I would avoid depending on a single arbitrary (α, β). The primary objective can instead be an uncertainty-aware balanced error:

$$J_t(\theta) = \frac{1}{2}U_{\mathrm{FP},t}(\theta) + \frac{1}{2}U_{\mathrm{FN},t}(\theta) + \lambda_{\mathrm{move}}\frac{|\theta-\theta_t|}{\sigma_s+\epsilon}.$$

Then report constrained operating points such as FNR at pre-registered FPR targets as secondary results.

Small-sample uncertainty

At $B = 16$, a class may have only a handful of queried defects. Treating an empirical rate from four or five defect anchors as exact will make the controller unstable.

The simplest near-term solution is to attach one-sided finite-sample intervals to candidate-threshold error counts. These intervals are not population guarantees when the labels were adaptively selected, but they are still useful **small-sample guards**. The paper must distinguish such empirical uncertainty guards from true anytime-valid guarantees.

Formal confidence-sequence claims become much harder once old anchors are repeatedly re-scored under a label-dependent memory state. Sun et al.’s theoretical guarantees apply to their own online thresholding model; they do not automatically transfer to a system that retroactively changes the score attached to previously observed examples whenever memory changes. [2] This is why the confidence-sequence candidate should not be first.

Selective-feedback bias

If BEA is deterministic, some samples have query propensity zero. In that case, IPW cannot reconstruct their labels.

A future propensity-aware controller therefore requires an exploration component satisfying

$$p_t = P(q_t = 1 \mid H_t, x_t) > 0.$$

Then a basic population-risk estimator is

$$\widehat{R}_{\mathrm{FP}}(\theta) = \frac{\sum_i \frac{q_i}{p_i}\mathbf{1}[y_i=0]\mathbf{1}[r_{i,t}\ge\theta]}{\sum_i \frac{q_i}{p_i}\mathbf{1}[y_i=0]},$$

with an analogous FNR estimator. Importance weighting is a standard way to correct active-sampling bias, but its variance can become large when propensities are small, which is why IWAL explicitly couples correction with variance control. [10]

Computational desiderata

Because DINOv2 is frozen, SCDAC should never repeatedly run the backbone merely to refresh threshold evidence. Cache frozen patch features once. The expensive online operation should be only native scoring of cached features against the current memory.

Let A be the number of calibration anchors, P patch count, K dynamic prototypes and d feature dimension. A naive current-state rescoring step is roughly

$$O(APKd)$$

for dense exact distance evaluation. In the main C64 protocol,

$$A \leq 64 + B \leq 192$$

before optional anchor pruning, so exact batched rescoring is a reasonable correctness-first implementation. Only after parity is established should it be optimized with incremental nearest-neighbor caches.

The paper should record scorer calls, bytes of anchor features, peak GPU memory and update latency. Calibration state is not “free memory.”

Candidate algorithms

Candidate A — State-Consistent Quantile Calibration

Priority: highest implementation priority; deliberately difficult to beat.

This is the simplest state-consistent method and should be treated as both a candidate and a falsification baseline.

Inputs are current Greedy memory M_t, 64 protected normal calibration anchors in the C64 protocol, optional leave-source-out queried-normal anchors, target normal tail probability α, maximum threshold movement Δ_θ, and current θ_t.

After any real memory change:

```
SCQ(M_t, ledger, theta_t):

    N <- eligible protected normal anchors

    if |N| is insufficient:
```

```
        return theta_t, NORMAL_EVIDENCE_INSUFFICIENT

    for each anchor i in N:
        score_i <- current_state_score(i, M_t)
        # leave-source-out if i contributed prototypes

    theta_raw <- empirical_quantile(score, 1 - alpha)

    theta_next <- clip(
        theta_raw,
        theta_t - Delta_theta,
        theta_t + Delta_theta
    )

    return theta_next, NORMAL_ONLY
```

The default paper baseline should be $\alpha = 0.10$, because it corresponds directly to the existing q90 calibration protocol. Sensitivity can use a pre-registered $\{0.05, 0.10, 0.20\}$ FPR grid.

Extreme imbalance: SCQ is unaffected by an absence of queried defects because it does not use them; consequently it must output `FNR_UNAVAILABLE`. That is a feature, not a bug.

Sidecar integration: replace the old normal-score buffer/fallback with `CalibrationLedger.current_normal_scores(memory_version)`. Existing threshold step clipping and cooldown can remain.

BEA integration: none initially. BEA is frozen out through Q-A.

Cost: one exact normal-anchor rescore per memory version; no iterative optimization.

Why it matters: if SCQ is already as good as SCDAC-G, the main claimed algorithmic novelty collapses to current-state recalibration, and we should accept that rather than embellish it.

Candidate B — Guarded State-Consistent Dual-Anchor Calibration

Priority: best near-term paper-centerpiece candidate.

SCDAC-G uses current-state normal and defect anchors and turns the existing Sidecar into an evidence-aware optimization layer.

Inputs:

$$(M_t, \theta_t, \mathcal{A}_N, \mathcal{A}_D, \alpha, \beta, \lambda_{\text{move}}, \Delta_\theta, \delta).$$

A practical version is:

```
SCDAC-G(M_t, ledger, theta_t):

    N <- current-state leave-source-out normal scores
    D <- current-state defect scores

    state <- classify_evidence(N, D)

    G <- sorted unique candidate thresholds
         from score midpoints + theta_t

    for theta in G:
        fpr_hat(theta) <- empirical normal error
        fnr_hat(theta) <- empirical defect error

        U_fpr(theta) <- small-sample upper bound if N available
        U_fnr(theta) <- small-sample upper bound if D available

    if state == BOTH:
        feasible <- {theta:
                     U_fpr(theta) <= alpha
                     and U_fnr(theta) <= beta}

        if feasible nonempty:
            theta* <- feasible theta minimizing
                      risk + movement penalty
            update last_certified_interval
        else:
            state <- CONFLICT
            theta* <- theta minimizing
                      0.5*U_fpr + 0.5*U_fnr
                      + movement penalty

    elif state == NORMAL_ONLY:
        # Never invent FNR evidence.
        theta* <- optimize normal side
                  only inside last certified defect-safe bound
        if no historical defect-safe bound:
            theta* <- hold theta_t

    elif state == DEFECT_ONLY:
        theta* <- optimize defect side
                  only inside last certified normal-safe bound
        if no normal-safe bound:
            theta* <- hold theta_t

    else:
        theta* <- theta_t
```

```
    theta_next <- bounded_move(theta_t, theta*, Delta_theta)

    return theta_next, state, diagnostics
```

I recommend two reporting modes.

The **primary BA mode** minimizes

$$\frac{1}{2}U_{\mathrm{FP}} + \frac{1}{2}U_{\mathrm{FN}} + \lambda_{\mathrm{move}}|\Delta\theta|.$$

The **risk-constrained mode** evaluates FNR under target FPR levels

$$\alpha \in \{0.05, 0.10, 0.20\}.$$

This avoids making the entire method depend on one arbitrary defect-risk target.

Extreme imbalance: preserve the last two-sided certified region and forbid evidence from one class from expanding the threshold in the direction that can silently damage the unobserved class.

Sidecar integration: this is the most direct evolution of the current controller. Keep candidate-threshold generation, move penalty, maximum step, cooldown and—if it survives ablation—the balanced-rescue mechanism. Replace scalar stale-score buffers with the versioned ledger and replace "missing class → zero rate" semantics with evidence flags.

Cost: rescoring $A_N + A_D$ anchors plus $O(G(A_N + A_D))$ CPU threshold evaluation; GPU cost is dominated by rescoring, not threshold search.

Candidate C — State-Consistent Dual-ACI

Priority: implement as a strong external-method adaptation, not assume it will become the new method.

ACI adapts a coverage parameter by realized error:

$$\alpha_{t+1} = \alpha_t + \gamma(\alpha - \mathrm{err}_t),$$

with γ controlling stability versus responsiveness. [16] Gibbs and Candès' 2024 extension dynamically tunes the effective step size to improve local adaptation. [5]

For our binary anomaly problem I would use **two class-conditional ACI states**, not one:

$$a_t^N, \qquad a_t^D.$$

```
SC-Dual-ACI(event, M_t, ledger):
```

```
    if memory version changed:
        rescore all eligible anchors under M_t

    theta_N <- Q_N(1 - a_N)
    theta_D <- Q_D(a_D) if defect evidence exists

    theta_t <- project previous theta
               into available [theta_N, theta_D] interval

    predict current image before seeing label

    if queried and y == NORMAL:
        e_fp <- 1[prediction == defect]
        a_N <- clip(a_N + gamma_N*(alpha_target - e_fp))

    if queried and y == DEFECT:
        e_fn <- 1[prediction == normal]
        a_D <- clip(a_D + gamma_D*(beta_target - e_fn))

    if one class was not observed:
        its ACI state does not update

    return theta_next
```

Use $\gamma = 0.005$ as the first fixed-step baseline because that is the illustrative experimental choice described in the original ACI paper, with a small pre-registered sensitivity around it rather than tuning against Probe A. [17]

This **does not inherit ACI's original theorem** under selective labels. ACI observes the realized error needed by its update; under BEA we observe only queried labels. That limitation must be explicit.

A second variant can adopt the JMLR 2024 adaptive step-size mechanism if ordinary ACI proves competitive. [5]

Candidate D — Confidence-Sequence SCDAC

Priority: theory-heavy second wave.

This is inspired by Sun et al.'s direct online anomaly-thresholding work. Their method uses confidence sequences to adapt thresholds under distribution shifts and supports warm-starting from relevant offline data. [2]

Our memory-aware version would construct uncertainty intervals around class-conditional risk or quantile estimates and update only when those intervals are informative:

```
CS-SCDAC(M_t, ledger, theta_t):

    rescore valid anchors under current memory

    build one-sided uncertainty sequence / interval
        for normal-tail risk
    build one-sided uncertainty sequence / interval
        for defect-miss risk, if valid defect evidence exists

    if both sides certify a nonempty safe threshold region:
        choose minimally-moving theta inside region

    elif current score lies in unresolved threshold band:
        expose "calibration uncertainty" to query controller

    elif evidence missing:
        hold / use last certified bound

    if a distribution-compatibility test fails:
        retire or downweight historical evidence
```

The difficulty is theoretical rather than computational. Re-scoring the same historical samples under a memory M_t that itself depends on past labels means the error sequence is no longer obviously the martingale/process assumed by an off-the-shelf confidence sequence.

A genuinely defensible CS version might therefore need **calibration epochs**: hold the memory fixed for a period, collect fresh randomly audited evidence under that scorer, then restart or safely warm-start the CS after a memory commit. That would materially change your current online-memory design.

For this reason I would not let CS-SCDAC delay the paper unless SCDAC-G first proves that the basic mechanism is worthwhile.

Candidate E — Propensity-Aware SCDAC

Priority: the right long-term answer to BEA bias, but not first implementation.

When the query policy is stochastic with known propensity p_i,

$$w_i = \frac{1}{p_i}$$

can correct selected-label risk estimates.

```
PA-SCDAC(event):

    p_t <- known query probability
```

```
    sample q_t ~ Bernoulli(p_t)

    if queried:
        record (anchor, y_t, p_t)

    after memory update:
        rescore anchors under current memory

    w_i <- min(1 / p_i, w_max)

    estimate propensity-weighted FPR/FNR
    compute effective sample sizes

    if ESS for either side below threshold:
        mark that risk unavailable

    choose theta using SCDAC-G objective
        with weighted risks and uncertainty
```

A deterministic BEA action with $p_t = 0$ on unqueried regions cannot support such correction. The query policy therefore needs an audit component, for example a small randomized exploration allocation.

Active, anytime-valid RCPS is especially relevant here because it explicitly studies adaptive label acquisition under a predetermined expected query fraction. [11] IWAL is the classical reference for bias correction via importance weighting. [10]

This extension becomes worthwhile only if fixed-Q-A experiments demonstrate that SCDAC itself is useful and later BEA experiments show a measurable selection-bias failure.

Candidate F — Covariate-Weighted SCDAC

Priority: lowest; only for residual environmental shift.

Current-state rescoring removes endogenous memory-state mismatch but cannot make historical anchors representative of a genuinely new production regime.

Let u_i be a frozen DINO global/CLS representation and let W_t be a recent unlabeled window. Estimate

$$\rho_i \approx \frac{p_t(u_i)}{p_{\text{anchor}}(u_i)}$$

using a simple density-ratio classifier or similarity kernel.

Then:

```
CW-SCDAC(M_t, ledger, recent_unlabeled):

    rescore calibration anchors under current M_t

    estimate anchor relevance weights rho_i
        from recent unlabeled features

    clip and normalize rho_i
    compute weighted class-conditional risks/quantiles

    if weight ESS collapses:
        return COVARIATE_EVIDENCE_INSUFFICIENT

    invoke SCDAC-G with weighted evidence
```

Weighted conformal work provides the conceptual precedent for likelihood-ratio weighting under covariate shift, but its validity depends on assumptions that may not hold under changing defect mechanisms or class-conditionals. [3] Therefore this should be an empirical robustness extension, not a claimed guarantee.

Candidate comparison

Candidate	Current-state rescore	Defect evidence	Missing-evidence semantics	Selection-bias correction	Environmental-shift handling	Theory potential	Near-term priority
SCQ	✓	no	✓	no	weak	low	**A**
SCDAC-G	✓	✓	✓	optional	weak	moderate	**A**
SC-Dual-ACI	✓	✓	partial	no initially	temporal adaptation	strong literature basis, adapted theorem unavailable	**A/B**
CS-SCDAC	✓	✓	✓	requires care	temporal shift	potentially high	**C**
PA-SCDAC	✓	✓	✓	**yes**	indirect	high if audit design is rigorous	**C**
CW-SCDAC	✓	✓	✓	optional	**yes**	assumption-sensitive	**D**

Evaluation and falsification protocol

The single most important experimental rule is:

> **Calibration algorithms must not be allowed to change the Greedy memory trajectory in the primary experiment.**

Otherwise a good threshold can change query choices or candidate patch counts, which changes memory, which changes future scores, and calibration attribution is lost.

Calibration-isolation protocol

For the primary causal experiment, run one shared scorer/memory trajectory:

$$M_0, M_1, \ldots, M_T,$$

and feed the same score state to every threshold controller.

The exact per-event protocol should be:

```
for event t:

    1. Compute native VisionAD score s_t using shared Greedy M_t.

    2. Every calibration method predicts using its own theta_t.
       No method sees y_t.

    3. Read the same frozen Q-A query decision for every method.

    4. If queried, reveal y_t to every calibration method simultaneously.

    5. Update the ONE shared Greedy memory using a
       threshold-independent, frozen candidate-write rule.

    6. If memory actually changed:
           increment memory_version
           rescore every eligible calibration anchor
           with source exclusion where required

    7. Each controller computes theta_(t+1)
       from exactly the same legal evidence.

    8. Log all decisions, scores, evidence states and hashes.
```

The memory writer should use a fixed, preregistered rule rather than each calibrator's own threshold. This avoids the circularity

$$\theta_t \to \text{memory write size} \to M_{t+1} \to s_{t+1} \to \theta_{t+1}.$$

Only after the best calibration algorithm is frozen should you conduct a fully closed-loop BEA experiment.

Information-access protocols

I recommend two protocols and absolutely no mixing of claims.

Protocol	Initial support	Extra normal calibration	Online queried labels	Purpose
Strict4	4	0	according to B	data-efficiency / reviewer fairness
C64	4	64 calibration-only normals	according to B	main reliable-calibration protocol

In C64, the 64 normals:

- are selected before the online stream;
- never enter the Greedy memory;
- never overlap evaluation sample groups;
- are re-scored under every new memory state;
- are identically available to every method allowed the C64 protocol.

This is consistent with the explicit P0/P1 data-access accounting, which records four-shot source support plus 64 extra training normals and no Probe-B access.

Experimental grid

The development grid should retain the already established axes:

Axis	Values
Query budget	$B \in \{16, 32, 64, 79, 128\}$
Support seeds	$0, 1, 2$
Q-A query seeds	$37, 101, 211$ where archived/registered
Greedy capacity	$64, 128, 256, 1024$
Stream order	Random/native, Normal-first, Defect-first, Bursty
Calibration access	Strict4, C64

Do **not** immediately run the full Cartesian product.

The main development configuration should use capacity 256 and budgets 32/64/79; $B = 16$ is the scarcity stress and $B = 128$ is the information-rich diagnostic. Capacities 64/128/1024 are mechanism stress tests after the controller is frozen.

Normal-first and Defect-first use labels to construct artificial diagnostic stream orderings. They are therefore **stress tests**, not claims about natural factory temporal processes. The controller still never sees those future labels during execution.

Primary and secondary metrics

For calibration-only comparisons, AUROC/AUPRC should be identical across threshold controllers because the memory trajectory and score sequence are shared. Any difference is a parity failure.

The main metric should therefore be prequential balanced error:

$$\mathrm{BER} = \frac{\mathrm{FPR} + \mathrm{FNR}}{2},$$

or equivalently Balanced Accuracy

$$\mathrm{BA} = 1 - \mathrm{BER}.$$

Pre-register:

Category	Metrics
Main operating performance	prequential FPR, FNR, BA
Ranking sanity	AUROC, AUPRC
Accumulated operational cost	cumulative FP, cumulative FN
Temporal robustness	worst defined-window FPR/FNR/BA
Evidence robustness	`BOTH/NORMAL_ONLY/DEFECT_ONLY/NONE/CONFLICT` fractions
State drift	stale-vs-current anchor score MAE, median, p95
Calibration dynamics	$\lvert\theta_t - \theta_{t-1}\rvert$, total variation, reset/hold count
Feasibility	dual-risk conflict fraction
Selection diagnostics	query class composition, propensity ESS where applicable
Compute	anchor-score calls, GPU peak memory, feature-cache bytes, p50/p95 update latency

For worst-window metrics, use the **official sample group** as the statistical unit. A practical preregistered definition is a rolling 32-group window, and define a class-conditional error rate only when at least five

observations of that class occur in the window. Otherwise mark the metric undefined and separately report the number of class-missing windows. This avoids turning “no defect occurred” into “FNR=0,” exactly the pathology visible in extreme temporal order experiments.

Calibration-drift statistics

Three new diagnostics should appear in the paper even if SCDAC eventually fails.

Memory-induced anchor drift

$$D_t^{\mathrm{mem}} = \mathrm{median}_{i \in \mathcal{A}_t} \left| \tilde{s}_{i,t} - s_{i,t_i} \right| .$$

Threshold stale-state error

$$D_t^{\theta} = \left| \theta_t^{\mathrm{stale}} - \theta_t^{\mathrm{current}} \right| .$$

Self-match bias

$$D_t^{\mathrm{self}} = \mathrm{median}_{i \in \mathcal{A}_N^{\mathrm{written}}} \left[S(x_i; M_t \setminus P_i) - S(x_i; M_t) \right] .$$

The third quantity tells us whether leave-source-out calibration is a real contribution or merely conceptual hygiene.

Candidate-to-baseline matrix

`P` means primary causal comparison; `S` means secondary system-level comparison; `D` means diagnostic/oracle only.

Candidate	Frozen θ	Rolling Q	SPOT	Greedy + current q90	Original Sidecar	ACI	Sun-CS	Causal Novelty	Novelty Top-K	R1 Beam replay
SCQ	P	P	P	same family	P	P	S	S	D	D
SCDAC-G	P	P	P	**P**	**P**	**P**	S/P if reproducible	S	D	D
SC-Dual-ACI	P	P	P	P	P	—	S	S	D	D
CS-SCDAC	P	P	**P**	P	P	**P**	**P**	S	D	D
PA-SCDAC	P	P	S	P	P	adapted propensity version	S	closed-loop S	D	D

Candidate	Frozen θ	Rolling Q	SPOT	Greedy + current q90	Original Sidecar	ACI	Sun-CS	Causal Novelty	Novelty Top-K	R1 Beam replay
CW-SCDAC	P	P	P	P	P	P	S	S	D	D

Novelty and Oracle must be paired with the **same calibrator** in system-level analyses. For example:

$$\text{Greedy} + \text{SCDAC-G} \quad \text{vs} \quad \text{Causal Novelty} + \text{SCDAC-G}.$$

Otherwise a memory-policy difference is confounded with a threshold-policy difference.

R1 Beam replay remains a hindsight diagnostic, not a deployable competitor. P0/P1 itself records that the oracle relies on Probe-A search labels.

Statistical analysis

Camera views from the same official sample group should not be treated as independent replicates. Use cluster bootstrap at sample-group level within a class/stream run, then macro-average over classes and seeds.

For the eventual paper, pre-register one primary contrast:

$$\Delta_{\text{BA}} = \text{BA}_{\text{SCDAC}} - \max_{\text{causal baselines}} \text{BA}_{\text{baseline}}.$$

The strongest causal baseline set should include at minimum:

$$\{\text{SCQ}, \text{ACI}, \text{SPOT}, \text{Original Sidecar}\}.$$

For multiple secondary pairwise tests, use either simultaneous bootstrap intervals or a Holm correction rather than selecting whichever comparator happens to give significance.

Provenance

Every run should write:

```
git_commit
protocol_hash
config_hash
dataset_manifest_hash
support_ids_hash
calibration_anchor_ids_hash
fixed_bank_hash
stream_hash
query_tape_hash
```

```
memory_writer_hash
memory_capacity
memory_version checkpoints
threshold-controller version
score-trace hash
output-file hashes
```

This follows the direction already established by P0/P1, whose archive contains dedicated provenance, query-tape and checksum artifacts.

Implementation roadmap and paper go/no-go

Repository architecture

I would not edit the 500k-line v22 script directly beyond a thin adapter. Put the new work behind a clean threshold-controller interface:

```
experiments/calibration/
    protocol.py
    runner.py
    feature_cache.py
    calibration_ledger.py
    state_rescorer.py
    evidence.py
    metrics.py
    provenance.py

    controllers/
        base.py
        frozen.py
        rolling_quantile.py
        spot.py
        original_sidecar_adapter.py
        scq.py
        scdac_guard.py
        dual_aci.py
        cs_scdac.py
        propensity_scdac.py
        covariate_scdac.py

tests/
    test_calibration_causality.py
    test_memory_versioning.py
    test_leave_source_out.py
    test_evidence_states.py
    test_extreme_imbalance.py
```

```
    test_shared_memory_parity.py
    test_aci_update.py
    test_propensity_weights.py
    test_provenance_hashes.py
```

The existing Sidecar should remain available as a frozen historical baseline.

Its new adapter can expose:

```python
class ThresholdController:
    def predict_threshold(self) -> float: ...
    def observe_label(self, event) -> None: ...
    def on_memory_commit(self, memory_state) -> None: ...
    def diagnostics(self) -> dict: ...
```

The key architectural change is that **buffers no longer own authoritative scores**. The ledger owns anchor identity and cached features; the state rescorer produces authoritative scores for the current memory version.

Required tests before any large GPU campaign

A few tests are more important than another experiment.

Causal-order test: modifying the current label must not change the current prediction/query decision.

State-invalidation test: changing memory version invalidates all cached current-state anchor scores.

Leave-source-out test: a normal anchor's own prototypes must be excluded during its calibration score.

Shared-memory parity test: Frozen, SCQ, ACI and SCDAC-G must produce exactly the same image scores and memory hashes under Q-A. Only thresholds/predictions may differ.

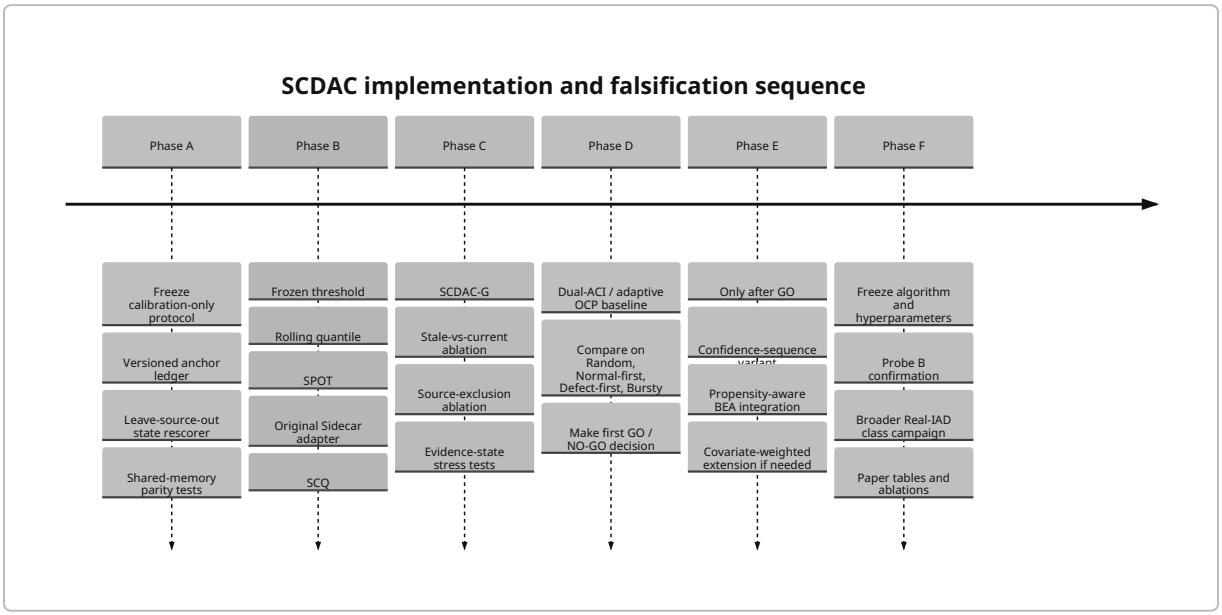
Missing-evidence test: a stream containing 1,000 normals and zero defects must never produce `FNR=0`; it must output `FNR_UNAVAILABLE`.

Conflict test: deliberately overlapping normal/defect scores must yield `CONFLICT`, not a false claim that both risk constraints are satisfied.

Probe-leakage test: Probe A/B paths cannot appear in threshold-controller inputs or calibration anchor manifests.

Extreme-order determinism test: Normal-first / Defect-first order manifests must hash reproducibly.

Prioritized build sequence



Rather than estimating wall-clock completion hours—which are dominated by cache and local GPU/IO behavior—I would budget the implementation by **development complexity and native-score workload**:

Component	Development burden	Incremental GPU workload	Build priority
Ledger + versioning	Medium	none by itself	immediate
LSO state rescorer	Medium	A anchor rescoring operations per memory commit	immediate
SCQ	Small	normal-anchor rescore only	immediate
SPOT adapter	Small	effectively none beyond score stream	immediate
Original Sidecar adapter	Small	none beyond existing controller	immediate
SCDAC-G	Medium	normal + defect anchor rescoring	immediate
Dual-ACI	Medium	same rescoring as SCDAC-G; $O(1)$ controller update	immediate
CS-SCDAC	Medium–large	similar scorer load; larger statistical engineering burden	conditional
PA-SCDAC	Large	multiple randomized closed-loop query runs	conditional

Component	Development burden	Incremental GPU workload	Build priority
CW-SCDAC	Large	anchor rescoring plus recent-window density-ratio computation	conditional

A single local smoke benchmark should convert those deterministic workloads into actual RTX 3090 Ti runtime measurements; those measured values can then be reported in the paper rather than guessed.

Strong stopping criteria

This project is especially vulnerable to method creep, so the stopping criteria should be frozen **before** tuning SCDAC.

State-consistency gate

SCDAC should not be a paper centerpiece unless current-state rescoring itself matters.

Compare identical controller logic using:

$$\text{stale anchor scores} \quad \text{vs} \quad \text{current-state rescored anchors.}$$

GO only if current-state rescoring yields either:

$$\Delta\text{BA} \geq 0.015$$

macro-averaged across development class/seed/order cells with a 95% cluster-bootstrap interval excluding zero, **or** a comparably clear reduction in preregistered worst-window error.

If there is no effect, “state consistency” is not strong enough to anchor the paper.

Simple-baseline gate

SCDAC-G must beat SCQ, not merely Frozen-θ.

A reasonable promotion criterion is:

$$\Delta\text{BA}_{\text{vs best simple}} \geq 0.02$$

with the 95% interval excluding zero across the development matrix.

It should also improve or be non-inferior on the worst stream order:

$$\Delta\text{BA}_{\text{worst order}} \geq -0.02.$$

If SCQ is statistically indistinguishable from SCDAC-G, use SCQ in the revised paper rather than invent extra complexity.

External-baseline gate

Before SCDAC becomes the first claimed innovation, it should beat at least:

- adapted ACI;
- SPOT;
- Rolling Quantile;
- Greedy + state-consistent q90;
- Original Sidecar,

under identical information access and shared memory.

Sun et al. is the strongest direct literature challenge and should be included if its algorithm can be faithfully reproduced under a clearly documented adaptation. Their paper directly claims adaptive anomaly thresholding and FPR/FNR guarantees in its own model, so omitting it from a threshold-centered paper would leave an obvious reviewer objection. [2]

Cross-class robustness gate

On the six current development classes, the candidate should show a positive primary-metric delta in at least **five of six classes** after seed aggregation, with no class suffering more than a 0.03 absolute BA degradation relative to the best causal comparator.

It should also retain a positive aggregate effect over support seeds rather than depending on one fortunate four-shot support draw.

Budget robustness gate

The expected pattern is allowed to be:

- $B = 16$: often evidence-insufficient and conservative;
- $B = 32$: partial adaptation;
- $B = 64/79$: principal operating regime;
- $B = 128$: diagnostic information-rich regime.

A candidate should not be promoted if its claimed advantage exists only at $B = 128$. The 128-label condition is useful scientifically but is not the same deployment regime as a sparse-feedback system.

Temporal-imbalance gate

On Normal-first, Defect-first and Bursty streams, the method should reduce worst-case operating-point collapse rather than merely improve the average.

In particular, promotion should require:

$$\mathrm{FNR}_{\mathrm{worst}}$$

not to increase by more than 0.05 relative to the strongest baseline and

$$\mathrm{BA}_{\mathrm{worst}}$$

to be non-inferior within 0.02.

The number of windows where each risk is unavailable should also be reported. Hiding undefined windows would defeat the purpose of evidence-aware calibration.

Novelty gate

Even a statistically successful controller may still be too incremental.

The paper-worthy contribution should empirically require **all three** of these components:

$$\boxed{\text{current-state rescoring} + \text{source-aware calibration} + \text{evidence-aware dual-sided control}}$$

If ablations show that ordinary current-memory q90 produces essentially all the gains, the honest contribution is a simpler calibration correction, not SCDAC.

Likewise, propensity-aware correction should not be added merely to make the algorithm look more sophisticated. It becomes necessary only if closed-loop BEA results show that queried-label calibration systematically diverges from full-stream offline evaluation.

Recommended first research campaign

The highest-information, lowest-risk sequence is:

$$\boxed{\text{Frozen } \theta \rightarrow \text{Rolling Q / SPOT} \rightarrow \text{SCQ} \rightarrow \text{Original Sidecar} \rightarrow \text{Dual-ACI} \rightarrow \text{SCDAC-G}}$$

all under one shared Greedy memory trajectory and one Q-A tape.

Only if SCDAC-G survives that comparison should the project branch into confidence sequences, propensity-aware BEA or covariate weighting.

That sequencing matters because the literature already gives us several plausible explanations for the problem: a simple streaming tail threshold may be enough; ACI-style correction may be enough; current-state q90 may be enough; or temporal selection bias may dominate. SPOT provides the first hypothesis, ACI/OCP the second, SCQ the third, and selective-label/active-risk literature the fourth. [18]

The central scientific question for the revised paper should therefore be narrower and more falsifiable than the previous system-level story:

> **When a frozen visual anomaly detector is adapted through a dynamic normal-memory bank, can calibration evidence be kept consistent with the current scoring state—and can an evidence-aware two-sided controller maintain a materially more reliable operating point than strong streaming and online-calibration baselines under sparse, temporally imbalanced feedback?**

That is a question with external baselines, a clear mechanism, a causal evaluation protocol, strong negative controls, and explicit stopping conditions. It is considerably safer as a paper direction than another memory-admission controller because **the underlying failure—threshold evidence becoming stale as Greedy memory changes—has already emerged directly from your recent experiments, while the proposed method can be falsified quickly by SCQ, ACI and SPOT rather than requiring another long chain of bespoke mechanisms.**

---

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